

Script-Controlled ACL – Restrict Record Access Based on Field Value

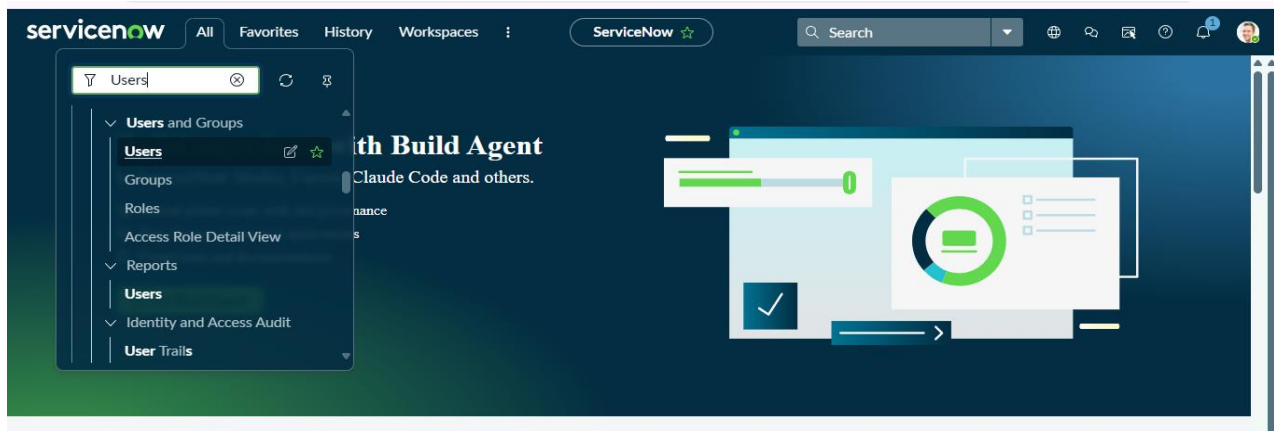
Description :

This lab demonstrates how to implement a script-controlled Access Control List (ACL) to restrict users from viewing records unless they meet specific conditions. In this scenario, only users belonging to the EEE branch are allowed to view records where the Branch field is set to EEE, while administrators retain full access.

Milestone-1 Creation of Users and Roles

User Creation:

1. **Log in** to the ServiceNow instance with **admin access**.
2. **Create a test user:**
 - Navigate to **User Administration** → **Users**
 - Click **New**
 - **User ID:** EEE User
 - **First name:** EEE
 - **Last name:** User
 - **Email:** eeeuser@gmail.com
 - Save the record



 This screenshot displays the 'User - EEE User' profile page in ServiceNow. The page is divided into two main sections: user details on the left and user preferences on the right. The left section includes fields for User ID (EEE User), First name (EEE), Last name (User), Title, Department, Password needs reset, Locked out, Active (checked), Identity type (Human), and Internal Integration User. The right section includes fields for Email (eeeuser@gmail.com), Language (None), Calendar integration (Outlook), Time zone (System (America/Los_Angeles)), Date format (System (2026-08-19)), Business phone, Mobile phone, and Photo (Click to add...). At the bottom, there are buttons for 'Update', 'Set Password', and 'Delete', along with 'Related Links' for 'View linked accounts' and 'View Subscriptions'.

Role Creation:

- Navigate to **User Administration** → **Roles**
- Click **New**
- **Name:** Enter role name as **bb1**
- **Description:** Brief role purpose
- Click **Submit**
- Create other roles also like **bb2**, **bb3** and **bb4**. (follow the same procedure)
- Assign all the roles to the **EEE User**

servicenow All Favorites History Workspaces User - EEE User

Roles

- Identity and Access Audit
 - Role Trails
 - User Role History
- User Administration
 - Time-Limited User Roles
 - Roles**
 - Role Delegation
 - Role Delegators
 - Delegate Roles in Group

Email eeeuser@gmail.com

Language -- None --

Calendar integration Outlook

Time zone System (America/Los_Angeles)

Date format System (2026-08-19)

Business phone

Mobile phone

Photo Click to add...

Identity type Human

Internal Integration User ☐

Update Set Password Delete

Related Links

[View linked accounts](#)

servicenow All Favorites History Workspaces Role - bb1

Name bb1

Application Global

Elevated privilege ☐

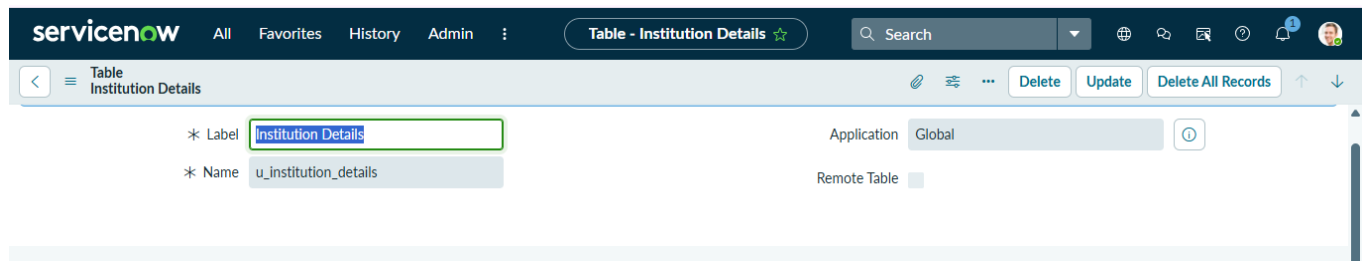
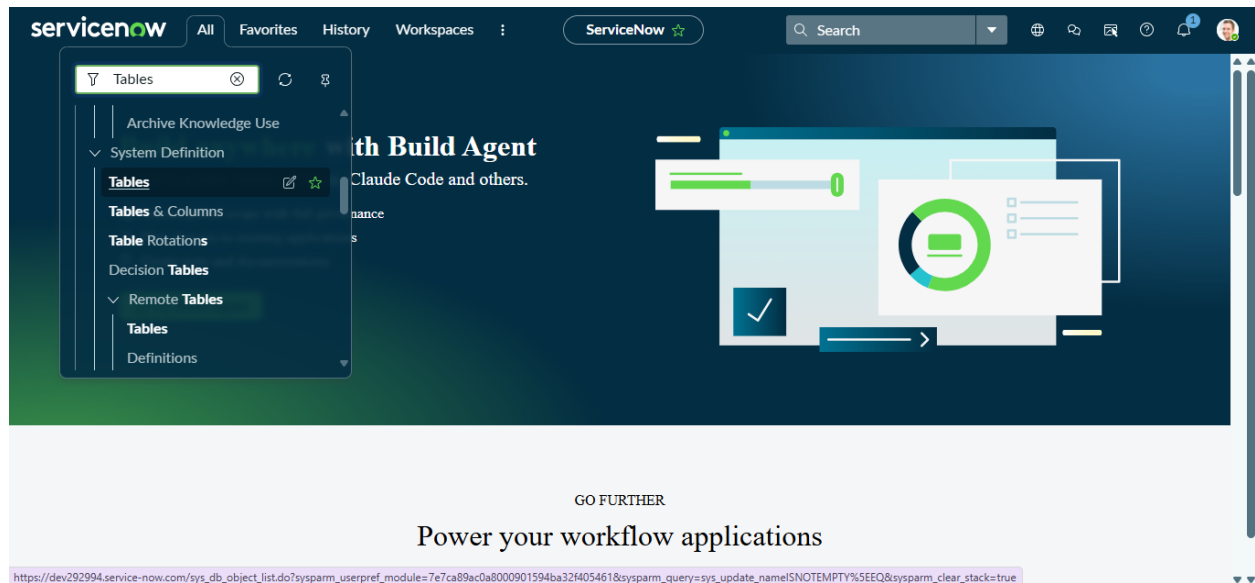
Description Practise

Update Delete

Milestone-2 Table creation

Tables Creation:

1. In the **Application Navigator**, search for **System Definition** → **Tables**.
2. Click **New** and create a custom table with the following details:
 - **Label:** Institution Details
 - **Name:** u_institution_details
 - **Extends for:** false



3. Click **Submit**.
4. Add the following fields to the **u_institution_details** table:
 - **Student Roll Number**– Auto Number
 - **Student Name** – Reference- User
 - **Faculty Name** – Reference- User
 - **Branch** – Choice (ECE, EEE, CSE)
 - **Email**– String
 - **Phone Number**– String
 - **Description**– Multi String
5. Save the Table.

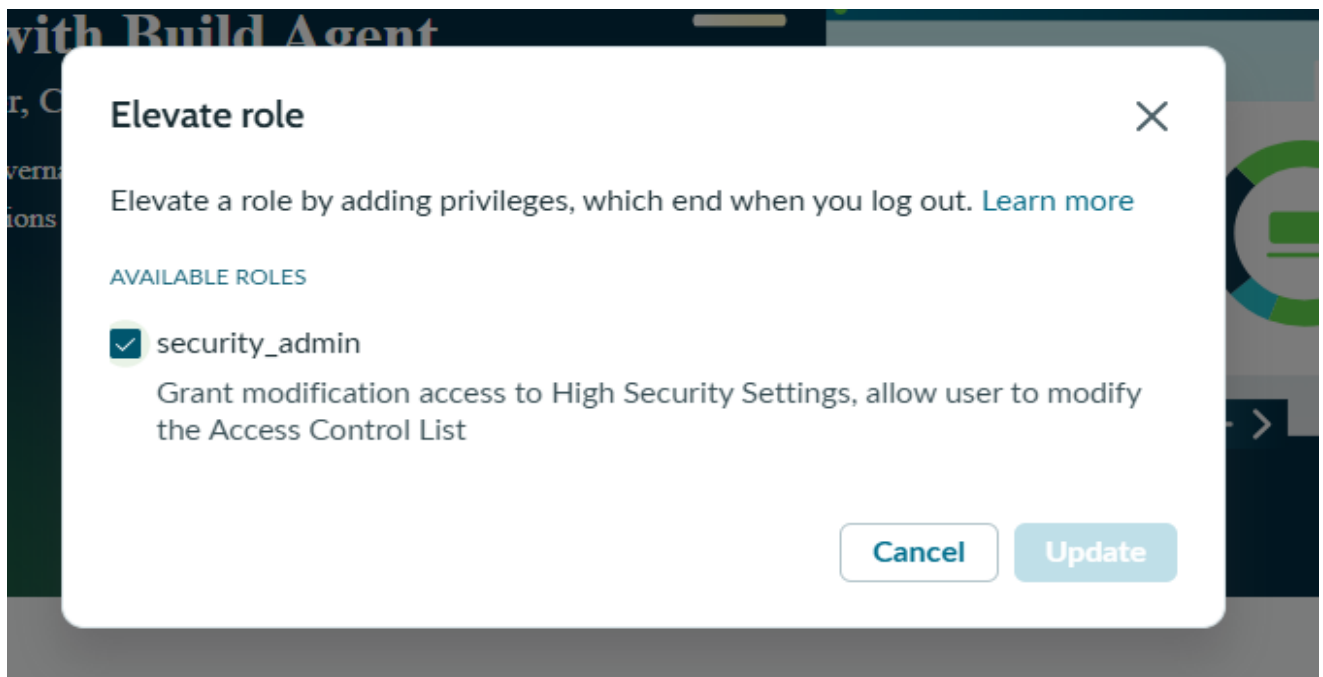
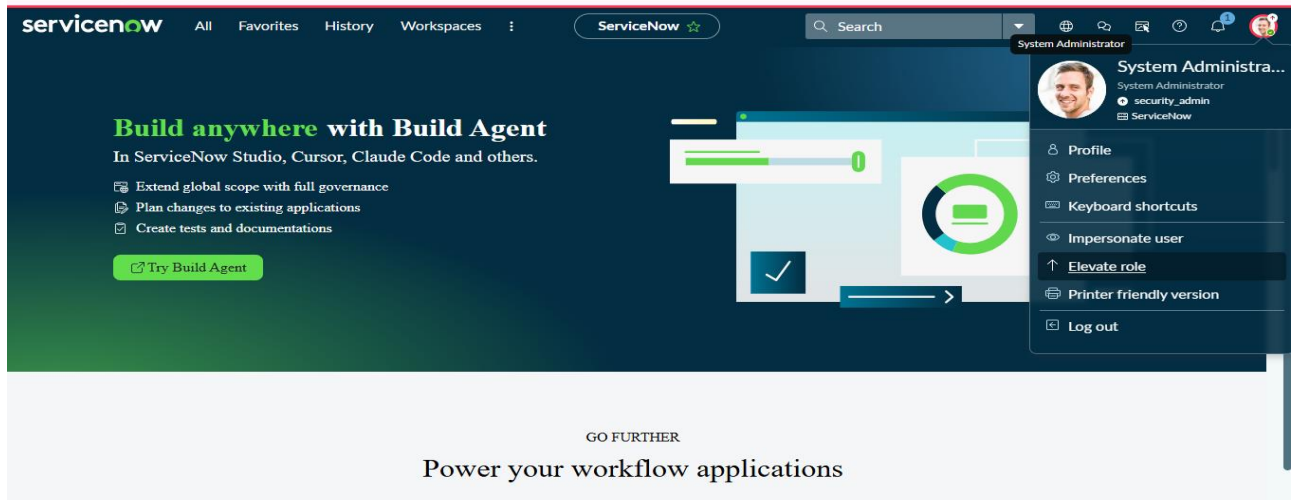
Milestone-3 Creation of Access Control Lists (Read ACL)

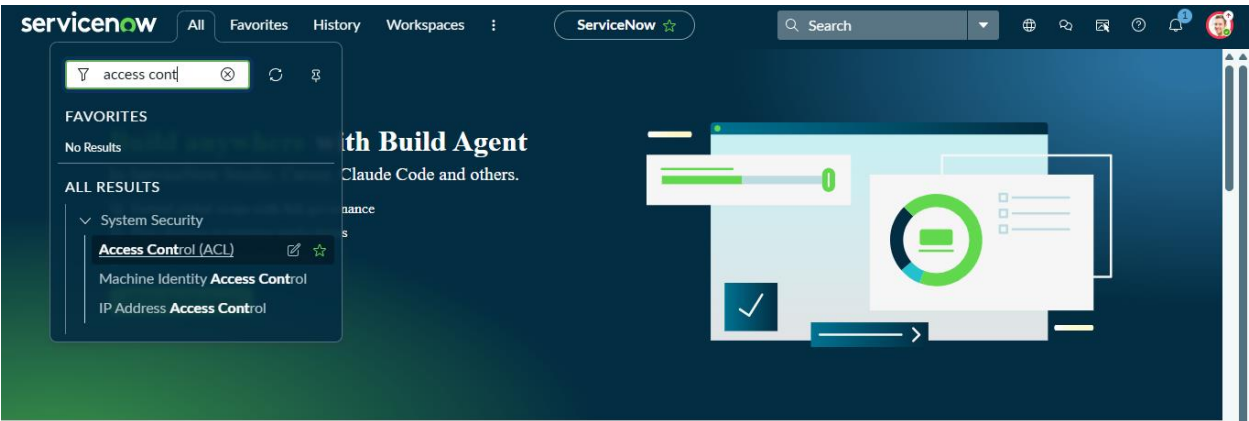
Creation of ACL – READ:

1. In the **Application Navigator**, search for **System Security → Access Control (ACL)**.
2. Elevate “Security_admin” role
3. Click **New** to create a new ACL.
4. Enter the following details:
 - **Type:** record
 - **Operation:** read
 - **Name:** u_institution_details
 - **Active:** true
5. In the **Requires role** related list, add:
 - bb1 (custom role)
6. Add Data condition as **Branch is EEE**
7. In the **Script** field, add the following script:

```
(function () {  
  
    // Allow admin users full access  
    if (gs.hasRole('admin')) {  
        return true;  
    }  
    // Allow only EEE branch users to see EEE records  
    if (gs.hasRole('bb1')){  
        return true;  
    }  
    // Deny access for all others  
    return false;  
})();
```

8. Click **Submit** to save the ACL.





GO FURTHER

Power your workflow applications

Access Controls							
Updated							
Search							
Actions on selected rows...							
New							
All							
☐	Name	Decision Type	Operation	Type	Active	Updated by	Updated
☐	u_institution_details	Allow If	delete	record	true	admin	2026-09-24 04:30:53
☐	u_institution_details	Allow If	write	record	true	admin	2026-09-24 04:29:05
☐	u_institution_details	Allow If	create	record	true	admin	2026-09-24 04:26:32
☐	u_institution_details	Allow If	read	record	true	admin	2026-09-23 07:10:43
☐	u_institution_details	Allow If	delete	record	true	admin	2026-09-23 00:09:41
☐	u_institution_details	Allow If	read	record	true	admin	2026-09-23 00:09:40
☐	u_institution_details	Allow If	create	record	true	admin	2026-09-23 00:09:40
☐	u_institution_details	Allow If	write	record	true	admin	2026-09-23 00:09:40
☐	sn_build_agent_task_telemetry	Allow If	report_view	record	true	system	2026-09-22 09:29:48
☐	sn_build_agent_conversation_change_log	Deny Unless	read	record	true	system	2026-09-22 09:29:48
☐	sn_build_agent_task_telemetry	Deny Unless	delete	record	true	system	2026-09-22 09:29:48
☐	sn_build_agent_task_telemetry	Deny Unless	report_view	record	true	system	2026-09-22 09:29:48
☐	sn_build_agent_conversation	Deny Unless	read	record	true	system	2026-09-22 09:29:48
☐	sn_build_agent_event_telemetry	Deny Unless	read	record	true	system	2026-09-22 09:29:48
1 to 20 of 44,187							

Access Control - u_institution_details

UpdateDelete

* Type

record

* Operation

read

Admin overrides

☒

Decision Type

Allow If

Protection policy

-- None --

* Name

Institution Details [u_institution_details]

Description

Allow read for records in u_institution_details, for users with role bb1, and if the ACL script returns true, and if the ACL condition (u_branch=EEE^EQ) evaluates to true.

Application

Global

Active

☒

Advanced

☒

Requires role



1 to 1 of 1



Role



bb1



Insert a new row...

Data Condition

Condition

No. of records matching the condition: 1

Add Filter Condition

Add OR Clause

Branch

is

EEE

AND

OR

X

Advanced Condition

Controlled by References



Script



Turn on ECMAScript 2021 (ES12) mode



```
1 function () {  
2   // Allow admin users full access  
3   if (gs.hasRole('admin')) {  
4     return true;  
5   }  
6   // Allow only EEE branch users to see EEE records  
7   if (gs.hasRole('bb1')){  
8     return true;  
9   }  
10  // Deny access for all others  
11  return false;  
12 }();  
13
```

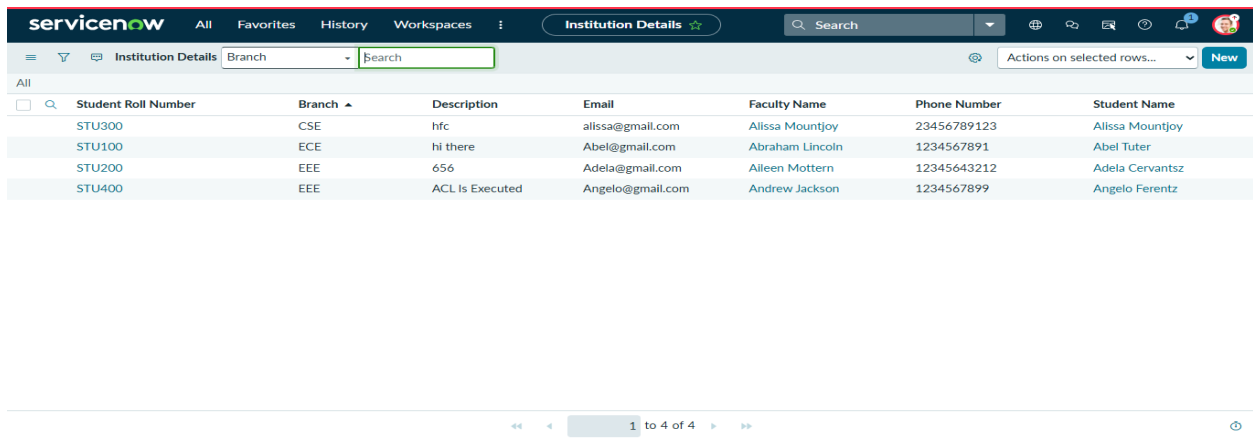
Update

Delete

9. Impersonate **EEE users** and open the Student Records list.
10. Copy the table name **u_institution_details.list** /LIST and paste in the All menu.
11. Impersonate **Admin** and open the Student Records list.

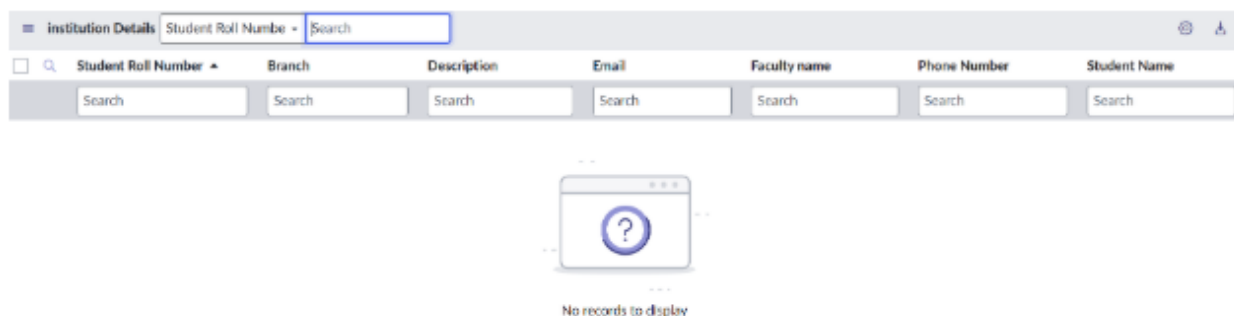
Verify:

- User with **bb1** role can view **only EEE branch records**.



Student Roll Number	Branch	Description	Email	Faculty Name	Phone Number	Student Name
STU300	CSE	hfc	alissa@gmail.com	Alissa Mountjoy	23456789123	Alissa Mountjoy
STU100	ECE	hi there	Abel@gmail.com	Abraham Lincoln	1234567891	Abel Tuter
STU200	EEE	656	Adela@gmail.com	Aileen Mottern	12345643212	Adela Cervantsz
STU400	EEE	ACL Is Executed	Angelo@gmail.com	Andrew Jackson	1234567899	Angelo Ferentz

- User without the role cannot view any records



- Admin user can view **all records regardless of branch**

servicenow							
All Favorites History Workspaces							
Institution Details							
Search							
Institution Details Branch Search							
Actions on selected rows... New							
All							
☐	Student Roll Number	Branch	Description	Email	Faculty Name	Phone Number	Student Name
	STU300	CSE	hfc	alissa@gmail.com	Alissa Mountjoy	23456789123	Alissa Mountjoy
	STU100	ECE	hi there	Abel@gmail.com	Abraham Lincoln	1234567891	Abel Tuter
	STU200	EEE	656	Adela@gmail.com	Aileen Mottern	12345643212	Adela Cervantsz
	STU400	EEE	ACL Is Executed	Angelo@gmail.com	Andrew Jackson	1234567899	Angelo Ferentz

Milestone-4 Creation of Access Control Lists (Create ACL)

Creation of ACL – CREATE:

1. Click **New** to create a new ACL(Create)
2. Enter the following details:
 - **Type:** record
 - **Operation:** create
 - **Name:** u_institution_details
 - **Active:** true
3. In the **Requires role** related list, add:
 - bb2 (custom role)
4. No need to add any Data condition
5. Save the ACL Record.

The screenshot shows the ServiceNow 'Access Control' console. The main configuration area includes the following fields:

- Type:** record
- Operation:** create
- Application:** Global
- Active:** ☒
- Advanced:** ☐
- Admin overrides:** ☒
- Decision Type:** Allow If
- Protection policy:** -- None --
- Name:** Institution Details [u_institution_details] (with a dropdown arrow)
- Description:** Allow create for records in u_institution_details, for users with role bb2.

Below the configuration area is a table titled 'Requires role' with the following content:

Role
bb2
Insert a new row...

Milestone-5 Creation of Access Control Lists (Write ACL)

Creation of ACL – WRITE:

1. Click **New** to create a new **ACL(Write)**
2. Enter the following details:
 - **Type:** record
 - **Operation:** Write
 - **Name:** u_institution_details
 - **Active:** true
3. In the Requires role related list, add:
 - bb3 (custom role)
4. No need to add any Data condition
5. Save the ACL Record.

servicenow All Favorites History **Access Control - u_institution_details** Search

Access Control u_institution_details Update Delete

* Type record Application Global

* Operation write Active ☒

Admin overrides ☒ Advanced ☐

Decision Type Allow If

Protection policy -- None --

* Name Institution Details [u_institution_details] -- None --

Description Allow write for records in u_institution_details, for users with role bb3.

Requires role 1 to 1 of 1

Role
X bb3
+ Insert a new row...

Verify :

- User with **bb1 & bb2 & bb3** roles can view and edit the **EEE branch records along with New Button**

servicenow All Favorites History Workspaces **Institution Details** Search

Institution Details Branch Search Actions on selected rows... New

All > Branch >= EEE

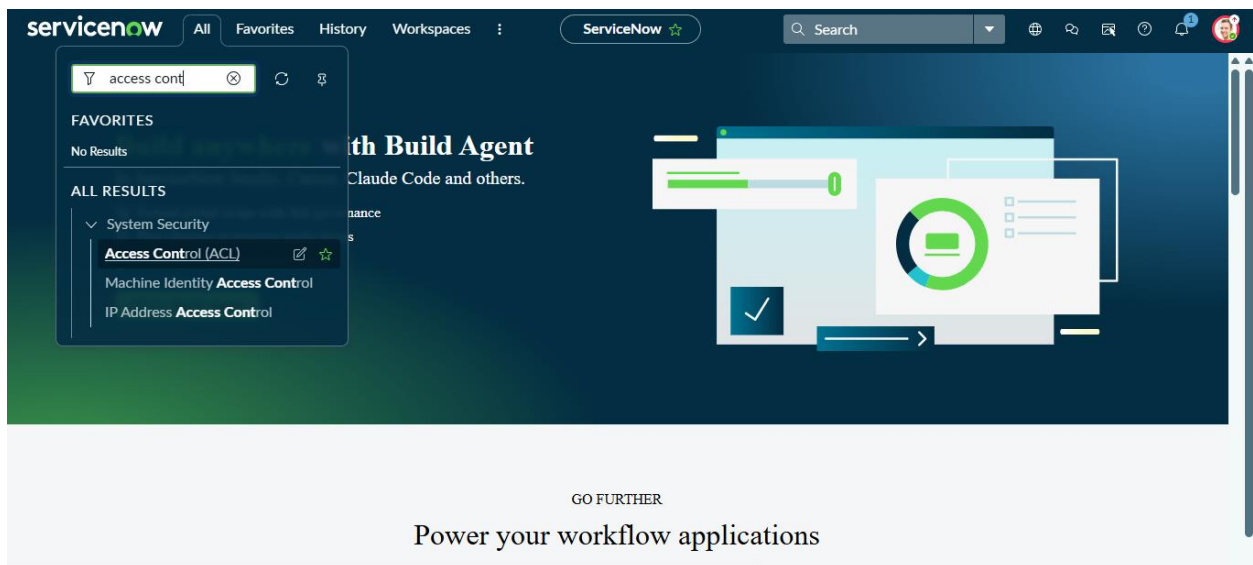
Student Roll Number	Branch	Description	Email	Faculty Name	Phone Number	Student Name
STU200	EEE	656	Adela@gmail.com	Aileen Mottern	12345643212	Adela Cervantsz
STU400	EEE	ACL Is Executed	Angelo@gmail.com	Andrew Jackson	1234567899	Angelo Ferentz

1 to 2 of 2

Milestone-6 Creation of Access Control Lists(Delete ACL)

Creation of ACL – DELETE:

1. Click **New** to create a new ACL(Delete)
2. Enter the following details:
 - **Type:** record
 - **Operation:** Delete
 - **Name:** u_institution_details
 - **Active:** true
3. In the **Requires role** related list, add:
 - bb4 (custom role)
4. No need to add any Data condition
5. Save the ACL Record.



servicenow

All

Favorites

History

Workspaces

Access Controls

Search

Access Controls

Updated

Search

Actions on selected rows...

New

All

☐	Q	Name	Decision Type	Operation	Type	Active	Updated by	Updated
		u_institution_details	Allow If	delete	record	true	admin	2026-09-24 04:30:53
		u_institution_details	Allow If	write	record	true	admin	2026-09-24 04:29:05
		u_institution_details	Allow If	create	record	true	admin	2026-09-24 04:26:32
		u_institution_details	Allow If	read	record	true	admin	2026-09-23 07:10:43
		u_institution_details	Allow If	delete	record	true	admin	2026-09-23 00:09:41
		u_institution_details	Allow If	read	record	true	admin	2026-09-23 00:09:40
		u_institution_details	Allow If	create	record	true	admin	2026-09-23 00:09:40
		u_institution_details	Allow If	write	record	true	admin	2026-09-23 00:09:40
		sn_build_agent_task_telemetry	Allow If	report_view	record	true	system	2026-09-22 09:29:48
		sn_build_agent_conversation_change_log	Deny Unless	read	record	true	system	2026-09-22 09:29:48
		sn_build_agent_task_telemetry	Deny Unless	delete	record	true	system	2026-09-22 09:29:48
		sn_build_agent_task_telemetry	Deny Unless	report_view	record	true	system	2026-09-22 09:29:48
		sn_build_agent_conversation	Deny Unless	read	record	true	system	2026-09-22 09:29:48
		sn_build_agent_event_telemetry	Deny Unless	read	record	true	system	2026-09-22 09:29:48

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servicenow

AllFavoritesHistory

Access Control - u_institution_details

Search

UpdateDelete

Access Controlu_institution_details

* Type: record

* Operation: delete

Admin override: ☒

Decision Type: Allow If

Protection policy: -- None --

* Name: Institution Details [u_institution_details]

Description: Allow delete for records in u_institution_details, for users with role bb4.

Application: Global

Active: ☒

Advanced: ☐

Requires role

Role
bb4
Insert a new row...

Security Attribute Condition

servicenow

AllFavoritesHistoryWorkspaces

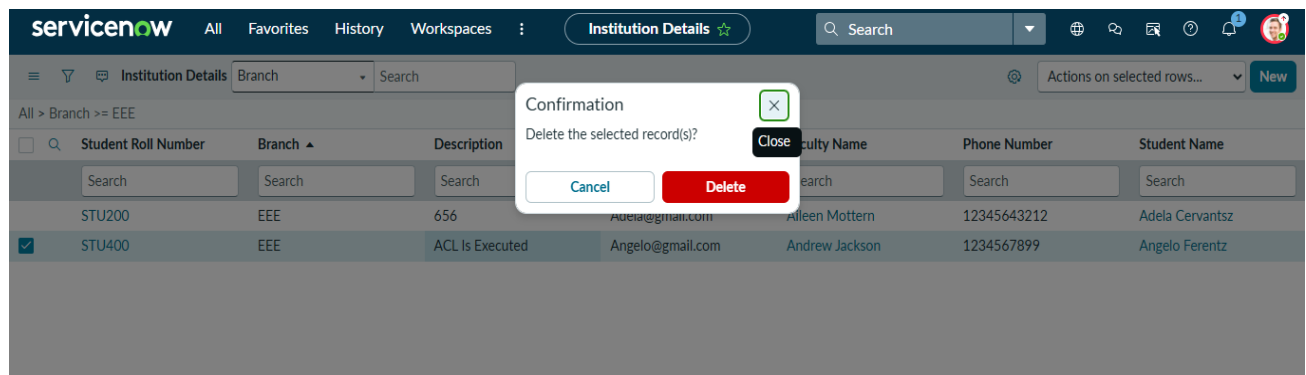
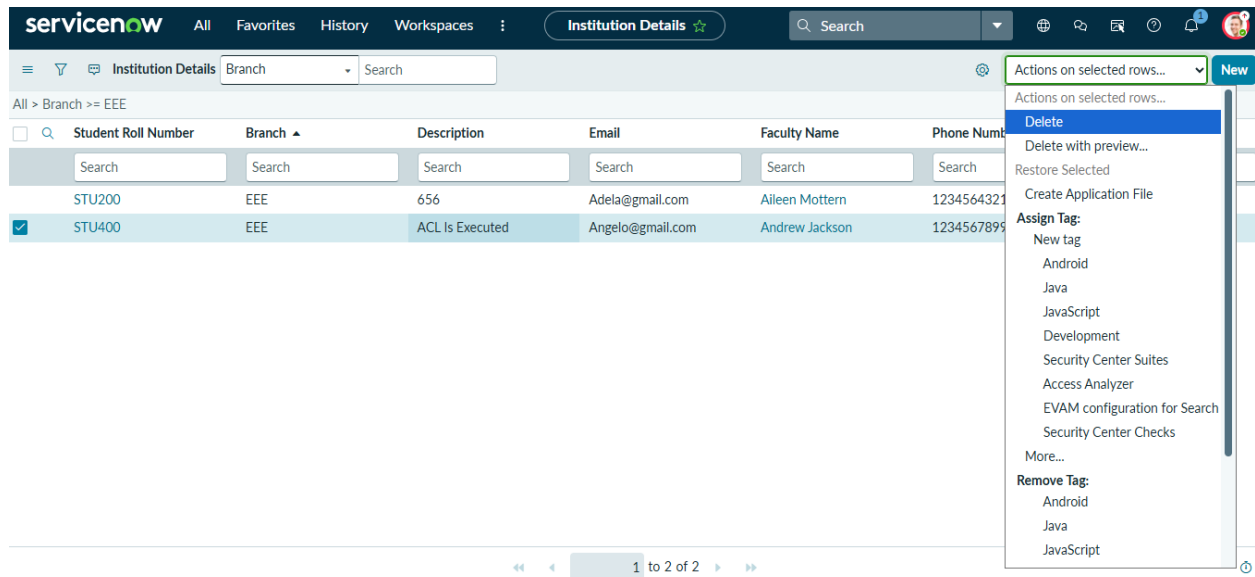
Institution Details

Search

Actions on selected rows...New

All > Branch >= EEE

Student Roll Number	Branch	Description	Email	Faculty Name	Phone Number	Student Name
STU200	EEE	656	Adela@gmail.com	Aileen Mottern	12345643212	Adela Cervantsz
STU400	EEE	ACL Is Executed	Angelo@gmail.com	Andrew Jackson	1234567899	Angelo Ferentz



OUTCOME:

By completing this lab, learners gain a comprehensive understanding of how script-controlled READ, WRITE, CREATE, and DELETE ACLs enforce record-level security, how user roles and record field values are evaluated together to make access decisions, and how ACLs protect sensitive data from unauthorized viewing, modification, creation, and deletion across forms, lists, and Service Portal interfaces.

